

Odious and evil integer partitions

Fixed order asymptotics, fixed multiplicity caps, and inverses

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Abstract

Positive integers are odious or evil according as their binary digit sum is odd or even. We prove the leading equivalents recorded as July 2025 conjectures in OEIS A067590 and A116492, derive the evil counterparts A067591 and A116491, and give expansions to every fixed algebraic order. A sectorial Mellin argument makes the signed part of the Euler product flat beyond its logarithm and constant. Uniform Fourier estimates exclude every competing arc, and a completed Hankel contour supplies a relative comparison with an explicit modified Bessel function. For every fixed multiplicity cap m , the odious-to-evil ratio is $m + 1$ up to every inverse power of the size. A specified smooth comparison has an all-order Lambert- W inverse; ordinary integer thresholds require rounding-aware enclosures. Established constant evaluations and analytic continuation are credited. No growing-cap uniformity, leading beyond-all-orders correction, numerical onset certificate, or historical-priority claim is made.

[write] **Status and trust boundary, added 6 October 2026, batch 107.** This is bundle Report 165 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a067590-odious-evil-partitions** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Research Report 165” and its PDF author field is empty. *What rests on what.* Theorems 1.1, 1.2 and 6.1 and Proposition 6.2 are proved by hand in Sections 2–6 from the Euler products, with classical inputs cited there (the Dedekind eta transformation, polynomial bounds for ζ in vertical strips, Hankel’s reciprocal-gamma integral and the modified Bessel series and expansions). The theorems are stated with $d = D'(0)$; Allouche’s evaluation (5) only rewrites the constants. No proof uses a computation or a decimal; the decimals of d , Q and $K_{\pm}$ are uncertified. In particular the two OEIS conjectures of Václav Kotěšovec (6 July 2025) are proved in their asymptotic form, with exact constants (Remark 1.3, which quotes them). The write adds three remarks with proofs: Remark 1.3, Remark 2.3 (an identity linking $D(s)$ to the Thue–Morse atlas’s shifted series $D(s, \frac{1}{2})$, a second route to $D(0) = -1$ and $D(-k) = 0$ through statements that are formal in Lean there, and exactly what is and is not formal) and Remark 6.3 (which parts of Section 6 are instances of the transseries volume’s theorems); and Section 10, the open questions, with a proof that the cap ratio (14) cannot hold uniformly in $1 \leq m \leq n$.

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1 Definitions and the principal results

Let $s_2(k)$ denote the number of ones in the binary expansion of k , and put

$$\varepsilon(k) = (-1)^{s_2(k)}, \quad \varepsilon(0) = 1, \quad E_\sigma = \{k \geq 1 : \varepsilon(k) = \sigma\}, \quad \sigma \in \{-1, +1\}.$$

The sign -1 denotes odious parts and $+1$ denotes evil parts. Only positive integers are allowed as parts: the evil integer zero is excluded. Write $p_\sigma(n)$ for the number of unrestricted partitions of n with parts in E_σ , and $p_{\sigma,m}(n)$ for the count when each part may occur at most m times. The empty partition gives all these sequences value 1 at $n = 0$. Throughout, m is a fixed positive integer. Every order of expansion and every closed acute sector is also fixed before the limiting variable tends to its endpoint.

For $\Re w > 0$ their convergent Euler products are

$$F_\sigma(w) = \sum_{n \geq 0} p_\sigma(n) e^{-nw} = \prod_{k \in E_\sigma} (1 - e^{-kw})^{-1}, \quad (1)$$

$$F_{\sigma,m}(w) = \sum_{n \geq 0} p_{\sigma,m}(n) e^{-nw} = \prod_{k \in E_\sigma} \sum_{j=0}^m e^{-jkw} = \frac{F_\sigma(w)}{F_\sigma((m+1)w)}. \quad (2)$$

All complex powers below use the branch positive on the positive real axis. Define

$$A = \frac{\pi^2}{12}, \quad \beta_\sigma = \frac{1}{4} + \frac{\sigma}{2}, \quad D(s) = \sum_{k \geq 1} \frac{\varepsilon(k)}{k^s} \quad (\Re s > 1), \quad (3)$$

$$d = D'(0), \quad C_\sigma = (2\pi)^{-1/4} e^{\sigma d/2}, \quad K_\sigma = \frac{C_\sigma A^{\beta_\sigma/2+1/4}}{2\sqrt{\pi}}. \quad (4)$$

The entire continuation defining $D'(0)$ is established work of Allouche–Cohen [7], discussed and used explicitly by Allouche; the estimates needed here are reproved in Section 2. With γ the Euler–Mascheroni constant and φ the Flajolet–Martin constant [8], the established evaluation is

$$d = -\log Q, \quad Q = \prod_{k \geq 1} \left(\frac{2k}{2k+1} \right)^{\varepsilon(k)} = \frac{e^\gamma}{\sqrt{2}\varphi}. \quad (5)$$

These identities appear in the proof of Allouche's Theorem 3.1 [5]; they are not new constants of this report. Define, for fixed real β , $a > 0$, and integer $j \geq 0$,

$$B_0(\beta, a) = 1, \quad B_j(\beta, a) = \frac{(-1)^j \prod_{r=1}^j (4(\beta+1)^2 - (2r-1)^2)}{j! (16\sqrt{a})^j}. \quad (6)$$

Theorem 1.1 (Unrestricted partitions). *For either sign, set $N = n - 1/48$. For each fixed integer $J \geq 1$,*

$$p_\sigma(n) = K_\sigma N^{-\beta_\sigma/2-3/4} e^{2\sqrt{AN}} \left\{ \sum_{j=0}^{J-1} B_j(\beta_\sigma, A) N^{-j/2} + \mathcal{O}_J(N^{-J/2}) \right\}. \quad (7)$$

More strongly, for every fixed real $R > 0$,

$$p_\sigma(n) = C_\sigma \left(\frac{A}{N} \right)^{(\beta_\sigma+1)/2} I_{-\beta_\sigma-1}(2\sqrt{AN}) \{1 + \mathcal{O}_R(n^{-R})\}. \quad (8)$$

Here I_ν is the modified Bessel function of the first kind.

In particular, the leading equivalents are

$$p_-(n) \sim K_- n^{-5/8} \exp\left(\pi\sqrt{n/3}\right), \quad (9)$$

$$p_+(n) \sim K_+ n^{-9/8} \exp\left(\pi\sqrt{n/3}\right). \quad (10)$$

The first is the equivalent conjectured by Václav Kotěšovec on 6 July 2025 in A067590 [1]; the second concerns A067591 [2]. For orientation only, non-interval-certified numerical evaluations give

$$\begin{aligned} d &\approx -0.4874506225215474, & Q &\approx 1.6281601297189172, \\ K_- &\approx 0.2218648339570363, & K_+ &\approx 0.1235806870073263. \end{aligned}$$

Keeping n unshifted, the first relative corrections are respectively

$$-\left(\frac{5\sqrt{3}}{32\pi} + \frac{\pi}{96\sqrt{3}}\right) n^{-1/2}, \quad -\left(\frac{45\sqrt{3}}{32\pi} + \frac{\pi}{96\sqrt{3}}\right) n^{-1/2}. \quad (11)$$

The exponential shift produces the second summand in each bracket; shifting the algebraic prefactor first contributes at order n^{-1} .

Theorem 1.2 (Every fixed multiplicity cap). *Fix $m \geq 1$, and put*

$$r = m + 1, \quad a_m = A \frac{m}{m+1}, \quad N = n + \frac{m}{48}, \quad K_{\sigma,m} = \frac{r^{-\beta_\sigma} a_m^{1/4}}{2\sqrt{\pi}}.$$

For each fixed $J \geq 1$,

$$p_{\sigma,m}(n) = K_{\sigma,m} N^{-3/4} e^{2\sqrt{a_m N}} \left\{ \sum_{j=0}^{J-1} B_j(0, a_m) N^{-j/2} + \mathcal{O}_{m,J}(N^{-J/2}) \right\}. \quad (12)$$

For every fixed $R > 0$ there is the relative comparison

$$p_{\sigma,m}(n) = r^{-\beta_\sigma} \left(\frac{a_m}{N} \right)^{1/2} I_{-1}(2\sqrt{a_m N}) \{1 + \mathcal{O}_{m,R}(n^{-R})\}. \quad (13)$$

Consequently,

$$\frac{p_{-,m}(n)}{p_{+,m}(n)} = m + 1 + \mathcal{O}_{m,R}(n^{-R}) \quad \text{for every fixed } R > 0. \quad (14)$$

No estimate here is asserted to be uniform when m grows with n .

At $m = 1$, these results give the distinct-part equivalents

$$p_{-,1}(n) \sim \frac{1}{2 \cdot 12^{1/4}} n^{-3/4} e^{\pi \sqrt{n/6}}, \quad (15)$$

$$p_{+,1}(n) \sim \frac{1}{4 \cdot 12^{1/4}} n^{-3/4} e^{\pi \sqrt{n/6}}, \quad (16)$$

The ratio is $2 + \mathcal{O}_R(n^{-R})$ for every fixed R . The odious leading constant is approximately 0.2686424829558855; this identifies the decimal in the July 2025 conjecture in A116492 [3]; the evil count is A116491 [4]. Equation (14) is a relative statement. It does not assert that the absolute difference $p_{-,m} - (m+1)p_{+,m}$ is polynomially small. We do not determine the leading residual beyond all algebraic orders.

Remark 1.3 ([write] Kotěšovec’s two conjectures, 6 October 2026). The live OEIS entries, read on 6 October 2026, are A067590 at revision #17 (July 6, 2025) and A116492 at revision #18 (July 6, 2025), the revisions the source inspected (Section 7). Their names are “Number of partitions of n into odious numbers (A000069).” and “Number of partitions of n into distinct odious numbers.”, so their terms are $p_{-}(n)$ and $p_{-,1}(n)$. Each carries one conjectural formula line:

Conjecture: $a(n) \sim c * \exp(\text{Pi} * \text{sqrt}(n/3)) / n^{(5/8)}$, where $c = 0.221864833\dots$ - _Vaclav
Kotesovec_, Jul 06 2025

in A067590, and

Conjecture: $a(n) \sim c * \exp(\text{Pi} * \text{sqrt}(n/6)) / n^{(3/4)}$, where $c = 0.26864248\dots$ - _Vaclav
Kotesovec_, Jul 06 2025

in A116492 (both quoted verbatim, including the four dots of the second). Equation (9) is the first with the exact constant

$$c = K_- = \frac{(2\pi)^{-1/4} Q^{1/2} (\pi^2/12)^{1/8}}{2\sqrt{\pi}}, \quad Q = \frac{e^\gamma}{\sqrt{2}\varphi},$$

and (15) is the second with $c = 1/(2 \cdot 12^{1/4})$; the exponents and exponentials agree, since $2\sqrt{An} = \pi\sqrt{n/3}$ and $2\sqrt{a_1 n} = \pi\sqrt{n/6}$. So both conjectures are proved as asymptotic equivalents. For the decimals: $1/(2 \cdot 12^{1/4}) = 0.2686424829558855\dots$ is an elementary closed form, and its first eight digits are those printed. The decimal $K_- \approx 0.2218648339570363$ comes from a floating-point evaluation of d ; the intake recomputed d independently from (25) (mpmath quadrature, -0.487450622521547) and obtained $K_- = 0.221864833957$, which begins with the nine printed digits. An independent check of the write (6 October 2026) enclosed $\log Q = \sum_{k \geq 1} \varepsilon(k) \log \frac{2k}{2k+1}$ in interval arithmetic (mpmath iv, 120 bits), summing Prouhet blocks of length 2^{10} (on which $\sum_{r < 2^{10}} \varepsilon(r) r^i = 0$ for $i < 10$) with the tail bound $|\text{block } n| \leq 1/(22n^{11})$, from Taylor’s theorem; it gives $d \in [-0.48745062252154755, -0.48745062252154734]$ and $K_- \in [0.22186483395703626, 0.22186483395703629]$. Given the established evaluation (5), the nine digits of c printed in A067590 are therefore correct (a machine interval computation, not a formal

proof). The quadrature (25) and the block sum, two independent mpmath routes, agree to 4×10^{-32} : $d = -0.4874506225215474445445025398\dots$, $Q = 1.6281601297189172488215312\dots$, $K_- = 0.22186483395703627316229215\dots$ and $K_+ = 0.12358068700732633070667746\dots$ (Added after the independent check of 6 October 2026; the first wording read “Neither evaluation is an interval enclosure, so that the digits of c printed in A067590 are right is supported by two concordant computations, not proved.”) The evil counterparts A067591 (revision #6) and A116491 (revision #14) carry no asymptotic formula. Both conjecture lines are still labelled conjectures in the OEIS; this is recorded here only, and nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 107.** *Provenance.* This report is Report 165 of the session bundle that ProveIt’s docs/incoming intake processed as batch 107: the archive `Odious_and_Evil_Partitions_All_Orders_and_Inverses_Source.zip` (669,797 bytes, SHA-256 `a0a70cc31baf...14b91ca`, 21 files at the archive root, no wrapper directory), arrival commit `60f54ea06`, placed in `3988bf5c4`. The text is the delivered `Report165.tex` (574 lines, dated 3 October 2026), shipped as `article.tex`. The delivered 15-page PDF and the two checksum manifests (`SHA256SUMS`, 20 entries, and `companion/reference_run/SHA256SUMS`, 2 entries; both verified at placement) are not shipped; the delivered README is replaced by the report README; all are retrievable from the arrival commit. The source’s pin is ProveIt commit `5dacd76e2` (3 October 2026), at which it inspected three archived Thue–Morse atlas files (Section 7); their blobs `c927e43a`, `7fee5be7` and `66c8012a` are unchanged at the commit of this write, so what the source says about them concerns their current text. The package carries no “prepared for private review” line and no personal data; it names Vladimir Reshetnikov only as the author of the cited atlas. It is a single-source report, so no merge choice was made. The write prefixed the 78 delivered labels with `oep:` and updated the 38 references to them; it added this subsection, the notes marked [write], Remarks 1.3, 2.3 and 6.3, Section 10 and a note in one bibliography entry. No statement, proof or number of the source was changed: the added remarks are the last statements of their sections, the added section comes after the last one, and the added displays are unnumbered, so every theorem, section and equation keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A067590, A067591, A116492, A116491 [1, 2, 3, 4], live entries at revisions #17, #6, #18 and #14, the revisions the source recorded. Their data sections (54, 63, 69 and 74 terms) agree term by term with the counts recomputed by the write from the definitions. The two conjecture lines are quoted in Remark 1.3.
- The current Thue–Morse atlas,
`Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Atlas_and_Frontiers/Thue_Morse_Atlas_and_Frontiers.tex`
(Part I: `p1:thm:evil-odious`, `p1:thm:boundary-flatness`, `p1:thm:Dirichlet-dyadic`,
`p1:thm:Mellin-integral`, `p1:cor:Dirichlet-trivial-zeros`, `p1:thm:G-functional`,
`p1:cor:G-Dirichlet`, `p1:thm:Woods-Robbins`, with their “Formal versions” paragraphs), and the Lean modules
`Analysis/FabiusFunction/Lean/FabiusFunction/ThueMorseBoundaryFlatness.lean`,
`ThueMorseEntireContinuation.lean`, `ThueMorseGDirichlet.lean`, `ThueMorseWoodsRobbins.lean`
and `ThueMorseEnumerators.lean` (declaration statements read; nothing was built). The archived atlas passages cited in Section 7 were spot-checked at their line numbers.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:def:model`,
`p0:thm:lambert-core`, `p0:thm:lambert-centered`, `p0:thm:core-reversion`,
`p0:thm:backward-error`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 6.3.
- Not read by the write: Allouche [5, 6], Tóth [9], Golafshan and Rigo [10], Merca [11], Allouche and Cohen [7], Flajolet and Martin [8], and the DLMF sections. The statements made about them in

Section 7 are the source's; `data/SOURCE_PROVENANCE.json` records what the source inspected, with SHA-256 fingerprints.

What was checked at intake. The batch-107 dossier read the manuscript in full and found no error; it rechecked by hand the pair bound (29) (maximum at $u = 1/\sqrt{3}$), the first corrections (11), the inverse constants of (60)–(62) and the model ratio $r^{-\beta_-}/r^{-\beta_+} = r$. Independently of the delivered code (own exact counts, mpmath at 25–40 digits) it found $d = -0.487450622521547$ from (25), $K_- = 0.221864833957$, $K_+ = 0.123580687007$, and $p_\sigma(n)/\mathcal{M}(n) - 1 = 3.33 \cdot 10^{-8}$ and $8.49 \cdot 10^{-10}$ at $n = 1000$ and 3000 for odious parts, $-4.40 \cdot 10^{-8}$ and $-7.77 \cdot 10^{-10}$ for evil parts (the Bessel model of (8)); the capped ratios at $n = 1500$ are 1.99999972 ($m = 1$) and 2.99999760 ($m = 2$); the four thresholds of Section 8 for the target 100000 were reproduced. The write rechecked the derivations of Sections 2–6 line by line and found no error, recomputed the live OEIS data, and reran the companion on copies (note at the end of Section 8). These numerical agreements are observations, not certificates.

Relation to the repository. Before batch 107 no file of the repository mentioned A067590, A067591, A116492 or A116491, or partitions into odious or evil parts. The Thue–Morse atlas in `Analysis/FabiusFunction/` overlaps in method and inputs only: the real-axis flatness of S and the continuation and zeros of the *shifted* series are there, formal in Lean, and Remark 2.3 says exactly which statements correspond, what is formal, and how the unshifted series used here relates to them. The atlas poses no question about partition counts with Thue–Morse-restricted parts, so this report does not continue it. The inverse construction of Section 6 and the transseries volume are compared in Remark 6.3. *Formal status:* no statement of this report is formalized in Lean or Rocq, and its place in the collection, or beside the formal Fabius development, confers no formal status.

[write] What is not claimed, collected from the source. Every order of expansion, every acute sector and every cap m is fixed: no estimate is uniform as m grows with n (Section 10 shows that full uniformity fails). The leading residual beyond all algebraic orders is not determined, and the relative statement (14) does not say that $p_{-,m} - (m+1)p_{+,m}$ is polynomially small. No constant $(d, Q, K_\pm)$, error constant, onset of eventual monotonicity or threshold constant $(C, y_0$ of Proposition 6.2) is interval-certified or evaluated; numerical quadrature supplies no enclosure for d (since the independent check of 6 October 2026, $\log Q$, and hence d and $K_\pm$ given (5), has an interval enclosure; Remark 1.3). A bare ceiling of an asymptotic inverse is not a threshold. No historical priority is claimed: the continuation of D and the evaluation of d are credited to Allouche–Cohen and Allouche; the literature search is bounded (Merca’s article read at abstract level only; the Allouche–Cohen and Flajolet–Martin originals not obtained; Golafshan–Rigo concern a different object), and its negative result is “not an exhaustive literature exclusion, a repository-wide originality certificate, or a historical-first claim”. The companion’s finite checks and the byte-identical rebuilds prove no asymptotic statement. No OEIS entry was edited. The write adds: the residual values quoted in Section 10 are floating-point observations and determine no rate.

Reading conventions (letters the manuscript reuses, and the neighbouring atlas's, with the tempting false readings; no symbol was renamed).

Symbol	Meaning here (where)	Not to be read as
σ	the sign ± 1 (-1 odious, $+1$ evil)	the atlas's $\sigma = \Re s$ in $D(\sigma, a)$
E_σ	allowed parts, positive integers only (zero, an evil number, is excluded)	the atlas's enumerators $\mathcal{E}_n$, $\mathcal{O}_n$ (from $\mathcal{E}_0 = 0$) or its product $\mathcal{E}(t)$
$S(w)$	$\sum_{k \geq 0} \varepsilon(k) e^{-kw} = \prod_j (1 - e^{-2^j w})$	(same function as the atlas's $\mathcal{E}(t)$ at real $w = t$)
$D(s), d$	$\sum_{k \geq 1} \varepsilon(k) k^{-s}$, unshifted; $d = D'(0)$; $D(0) = -1$	the atlas's shifted $D(s, a) = \sum_{n \geq 0} \varepsilon_n (n + a)^{-s}$, $a > 0$, with $D(0, a) = 0$ (Remark 2.3)
$G(s), J(s)$	$\Gamma(s)D(s)$; the Mellin integral (18)	the atlas's master product $G(a, b)$; the order J ; the block length 2^J (Section 4)
$H(w)$	the signed logarithm (22)	
Q, φ	$e^\gamma / (\sqrt{2}\varphi)$, φ the Flajolet–Martin constant	the decay power Q and the angle ϕ of Lemma 2.2
θ, t	a sector angle (Sections 2–3); a Fourier angle and $t = \Re w$ (Sections 4–5)	the Mellin variable t (Section 2); the formal variable t of (55)
N	$n - \frac{1}{48}$ (Theorem 1.1); $n + \frac{m}{48}$ (Theorem 1.2); $n + \lambda$ (Section 5)	the prefix length in (30)
A, a, a_m	$\pi^2/12$; the general exponent of (38); $Am/(m+1)$	
B, B_j, b	the pair bound $16/(3\sqrt{3})$; Bessel coefficients (6); $2\sqrt{a}$	the b of <code>p0:thm:lambert-core</code> , which is $-2p$ here (Remark 6.3)
p, p_σ, p_j	the exponent $\beta/2 + 3/4$; the counts; probabilities in Section 4	each other
C_σ, K_σ, K, C	model constants; C, y_0 of Proposition 6.2 are unrelated error constants	Kotěšovec's c is K_- resp. $K_{-,1}$
r, m	$m + 1$; the cap	r as radius (proof of Lemma 2.2) or product index in (6)
L, R, q	flatness orders; a fixed power; a derivative order (Lemma 2.1)	$q = e^{-t}$ in (32), the exponent q of Proposition 6.2; the target L of the transseries volume
δ, ℓ	the arc exponent; the summation index in (22)	$\delta_{<m}$, δ_J and ℓ_j of Theorem 6.1
$\mathcal{M}, T(y)$	the Bessel model; the first threshold (63)	$T(y)$ is the staircase $N_*(y)$ of <code>p0:def:three-inverses</code> only on the monotone tail

2 Sector flatness and the signed Mellin transform

The Thue–Morse recurrences $\varepsilon(2k) = \varepsilon(k)$ and $\varepsilon(2k+1) = -\varepsilon(k)$ give, by finite products and then normal convergence,

$$S(w) = \sum_{k \geq 0} \varepsilon(k) e^{-kw} = \prod_{j \geq 0} (1 - e^{-2^j w}), \quad \Re w > 0. \quad (17)$$

The finite product with $j < J$ is the Fourier–Laplace polynomial of the first 2^J signs. The infinite product and all its derivatives are holomorphic in the right half-plane.

Lemma 2.1 (Flatness with derivatives). *Fix $\theta < \pi/2$. For each real $L \geq 0$ and each fixed integer $q \geq 0$,*

$$S^{(q)}(w) = \mathcal{O}_{L,q,\theta}(|w|^L) \quad (w \rightarrow 0, |\arg w| \leq \theta).$$

As $|w| \rightarrow \infty$ in that sector, $S(w) - 1$ and every positive-order derivative are exponentially small.

Proof. Put $\rho = |w|$, $c = \cos \theta > 0$, and $k = \lfloor \log_2(1/\rho) \rfloor$ for small ρ . For $j < k$, the integral identity $1 - e^{-z} = z \int_0^1 e^{-uz} du$ gives $|1 - e^{-2^j w}| \leq 2^j \rho$. Since $2^k \rho > 1/2$, the tail is bounded by

$$\prod_{j \geq k} |1 - e^{-2^j w}| \leq \prod_{h \geq 0} (1 + e^{-c2^{h-1}}) = C_\theta < \infty.$$

Thus $|S(w)| \leq C_\theta \rho^k 2^{k(k-1)/2}$, and the logarithm of the right side is

$$-\frac{(\log(1/\rho))^2}{2 \log 2} + \mathcal{O}_\theta(\log(1/\rho)).$$

It is eventually below $L \log \rho$ up to a constant, for every prescribed L . To differentiate, take a slightly wider acute sector. A disk of radius $\eta|w|$ around each point in the original sector stays in the wider one, with moduli comparable to $|w|$. Cauchy's estimate applied to flatness of order $L + q$ proves the claim for $S^{(q)}$. At infinity, the absolutely convergent series and its differentiated series are bounded by exponentially decaying geometric sums. $\square$

Set

$$G(s) = \Gamma(s)D(s), \quad J(s) = \int_0^\infty t^s S'(t) dt. \quad (18)$$

Lemma 2.1 shows that J is entire, locally uniformly in s . Initially for $\Re s > 1$, Mellin integration and integration by parts give

$$G(s) = \int_0^\infty t^{s-1} (S(t) - 1) dt = -\frac{J(s)}{s}. \quad (19)$$

The first integral itself converges for $\Re s > 0$. The integration-by-parts boundary terms vanish there. Since $J(0) = S(\infty) - S(0) = 1$, G has its only pole at zero, with residue -1 . Dividing by $\Gamma(s)$, whose reciprocal is entire with simple zeros at the nonpositive integers, proves the continuation and the values

$$D(0) = -1, \quad D(-k) = 0 \quad (k = 1, 2, \dots). \quad (20)$$

This is a proof of an established continuation in the form needed here, not a priority claim. The shifted series $\sum_{k \geq 0} \varepsilon(k)/(k+1)^s$ appearing in several sources is different; its value at zero is 0, not -1 .

Lemma 2.2 (Vertical estimates). *For a fixed strip $u \leq \Re s \leq v$, angle $0 < \phi < \pi/2$, and every fixed $Q > 0$,*

$$|J(c + i\tau)| \leq C_{u,v,\phi,Q} e^{-\phi|\tau|} (1 + |\tau|)^{-Q}. \quad (21)$$

An estimate of the same form, after changing the polynomial power, holds for $G(s)$ away from its pole.

Proof. For $\tau \neq 0$ let $\eta = \operatorname{sgn} \tau$. Rotate the integration ray in (18) to angle $\eta\phi$. Flatness removes the small joining arc, and the exponential decay of S' removes the large one. With continuous powers in the right half-plane this gives

$$J(s) = e^{i\eta\phi(s+1)} \int_0^\infty r^s S'(re^{i\eta\phi}) dr.$$

After $r = e^x$, the integral is the Fourier transform in τ of $e^{(c+1)x} S'(e^{x+i\eta\phi})$. This function and each fixed number of its x -derivatives are integrable uniformly for $u \leq c \leq v$. At $-\infty$ use arbitrary-order derivative flatness; at $+\infty$ use faster-than-exponential decay in x . Repeated integration by parts gives arbitrary inverse powers of $1 + |\tau|$, while the prefactor has modulus $e^{-\phi|\tau|}$. Bounded τ follows directly from the convergent integral. Finally use (19). $\square$

The signed logarithm of the Euler product, specified by its absolutely convergent logarithmic series, is

$$H(w) = \sum_{k \geq 1} \varepsilon(k) \log(1 - e^{-kw})^{-1} = \sum_{\ell \geq 1} \frac{S(\ell w) - 1}{\ell}. \quad (22)$$

Mellin inversion, initially on a vertical line $\Re s = c > 1$, yields

$$H(w) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} G(s) \zeta(s+1) w^{-s} ds. \quad (23)$$

Absolute convergence justifies the initial interchange. By $D(0) = -1$, $D'(0) = d$, and the expansions of Γ and ζ ,

$$G(s) = -\frac{1}{s} + d + \gamma + \mathcal{O}(s), \quad G(s) \zeta(s+1) = -\frac{1}{s^2} + \frac{d}{s} + \mathcal{O}(1).$$

The residue after multiplication by w^{-s} is $\log w + d$. There are no other poles. Indeed G is holomorphic away from zero and $\zeta(s+1)$ has no other pole.

Fix $|\arg w| \leq \theta < \phi < \pi/2$. Classical polynomial bounds for ζ in fixed vertical strips, together with Lemma 2.2, bound the integrand on the horizontal edges by a polynomial times $e^{-(\phi-\theta)|\Im s|}$. The horizontal integrals vanish as their heights tend to infinity. Moving the contour to $\Re s = -L$ therefore proves, for every fixed $L > 0$,

$$H(w) = \log w + d + \mathcal{O}_{L,\theta}(|w|^L). \quad (24)$$

In particular, no negative-integer residues were omitted. Taking the constant term at $s = 0$ in the real integral (19) also gives the convergent formula

$$d = -\gamma + \int_0^1 \frac{S(t)}{t} dt + \int_1^\infty \frac{S(t) - 1}{t} dt. \quad (25)$$

This identity explains a convenient numerical evaluation, but numerical quadrature by itself supplies no certified enclosure for d .

Remark 2.3 ([write] Relation to the Thue–Morse atlas, 6 October 2026). The current atlas (path in Section 1.1) writes $\mathcal{E}(t) = \prod_{j \geq 0} (1 - e^{-2^j t})$, which is $S(t)$ at real $t > 0$, and, for $a > 0$, $D(s, a) = \sum_{n \geq 0} \varepsilon_n (n + a)^{-s}$, a *shifted* series ($\varepsilon_n = \varepsilon(n)$).

(a) *What corresponds.* At real w , Lemma 2.1 with $q = 0$ is the flatness `p1:eq:superpolynomial-flatness` of `p1:thm:boundary-flatness`; the atlas proves it, with the two-sided logarithmic asymptotic, for real t only, and the bound $|S(w)| \leq C_\theta \rho^k 2^{k(k-1)/2}$ above is the sector form of

the upper half of its proof. The atlas continues $D(s, a)$ for $a > 0$ by the Mellin kernel $e^{-at}\mathcal{E}(t)$ (`p1:thm:Mellin-integral`) and proves $D(-r, a) = 0$ for all integers $r \geq 0$ (`p1:cor:Dirichlet-trivial-zeros`). The unshifted $D(s)$ of this report is not a member of that family: its kernel is $S(t) - 1$ in (19), and its value at $s = 0$ is -1 , not 0 . The sector form of Lemma 2.1, its derivative bounds, Lemma 2.2 and (24) have no counterpart in the atlas.

(b) *A bridge.* For every $s \in \mathbb{C}$,

$$D(s, \tfrac{1}{2}) = (1 - 2^s) D(s).$$

Proof. For $\Re s > 1$ all series converge absolutely. Split $k \geq 1$ by parity: the even $k = 2j$, $j \geq 1$, contribute $\sum_{j \geq 1} \varepsilon(j)(2j)^{-s} = 2^{-s}D(s)$, since $\varepsilon(2j) = \varepsilon(j)$; the odd $k = 2j + 1$, $j \geq 0$, contribute $-\sum_{j \geq 0} \varepsilon(j)(2j+1)^{-s} = -2^{-s}D(s, \tfrac{1}{2})$, since $\varepsilon(2j+1) = -\varepsilon(j)$. Hence $(1 - 2^{-s})D(s) = -2^{-s}D(s, \tfrac{1}{2})$, and multiplying by -2^s gives the identity for $\Re s > 1$. Both sides are entire, the left by `p1:thm:Mellin-integral` and the right by (19)–(20), so the identity holds on $\mathbb{C}$. $\square$

Consequences: differentiating once and twice at $s = 0$ and using $D(0) = -1$, $D'(0) = d$,

$$D(0, \tfrac{1}{2}) = 0, \quad \partial_s D(s, \tfrac{1}{2})|_{s=0} = \log 2, \quad \partial_s^2 D(s, \tfrac{1}{2})|_{s=0} = \log^2 2 - 2d \log 2;$$

the first agrees with the atlas, and the third expresses the constant d of this report through the atlas's continuation. Moreover $D(2\pi i k / \log 2, \tfrac{1}{2}) = 0$ for every integer k , because D is entire. (Numerically, at the write: the Mellin value $\int_0^\infty e^{-t/2} S(t) dt/t$, which is the first derivative because $1/\Gamma(s) = s + \mathcal{O}(s^2)$, equals $\log 2$ to 20 digits, and the second derivative, computed from the Mellin integral, agrees with $\log^2 2 - 2d \log 2 = 1.156203063244203$ to 16 digits; `mpmath`, uncertified.)

(c) *A second route to (20).* Three atlas statements, with its zeros at $s = 0$, give the value $D'(0, \tfrac{1}{2}) = \log 2$ without this report. The dyadic equation $D(s, 1) = 2^{-s}(D(s, \tfrac{1}{2}) - D(s, 1))$ (`p1:thm:Dirichlet-dyadic` at $a = 1$, in its entire form `p1:eq:Dirichlet-dyadic-entire`, valid for every $s \in \mathbb{C}$ by the identity theorem and formal as `Fabius.dirichletMellinContinuation_dyadic`; the theorem itself is stated for $\Re s > 0$ only) (entire form cited after the independent check of 6 October 2026; the first wording cited only “`p1:thm:Dirichlet-dyadic` at $a = 1$ ”), differentiated at $s = 0$ with $D(0, \tfrac{1}{2}) = D(0, 1) = 0$, gives $D'(0, 1) = D'(0, \tfrac{1}{2}) - D'(0, 1)$. The derivative bridge $\log G(a, b) = D'(0, b) - D'(0, a)$ (`p1:cor:G-Dirichlet`) at $(a, b) = (\tfrac{1}{2}, 1)$, with the Woods–Robbins product $G(\tfrac{1}{2}, 1) = \prod_{n \geq 0} ((2n+1)/(2n+2))^{\varepsilon_n} = 1/\sqrt{2}$ (`p1:thm:Woods-Robbins`), gives $D'(0, 1) - D'(0, \tfrac{1}{2}) = -\tfrac{1}{2} \log 2$. Hence $D'(0, 1) = \tfrac{1}{2} \log 2$ and $D'(0, \tfrac{1}{2}) = \log 2$. By the identity (b) on $\Re s > 1$ alone, $D(s, \tfrac{1}{2})/(1 - 2^s)$ is the meromorphic continuation of D ; letting $s \rightarrow 0$ gives $D(0) = D'(0, \tfrac{1}{2})/(-\log 2) = -1$. Likewise $D(-k) = D(-k, \tfrac{1}{2})/(1 - 2^{-k}) = 0$ for $k \geq 1$ by `p1:cor:Dirichlet-trivial-zeros`. This reproves both values of (20) from the atlas and the elementary identity (b); it does not touch d , which (b) only relates to the second derivative above. The write did not find (b) or the value $D'(0, \tfrac{1}{2}) = \log 2$ stated in the atlas; both are elementary, and no novelty is claimed.

(d) *Formal status.* Formal in Lean (statements read at the commit of this write): the effective real-axis bounds $2^{\binom{m}{2}}(t/2)^m \mathcal{E}(\tfrac{1}{2}) \leq \mathcal{E}(t)$ when $\tfrac{1}{2} \leq 2^m t \leq 2$, and $\mathcal{E}(t) \leq 2^{\binom{m}{2}} t^m$ for all $t > 0$, $m \geq 0$ (`Fabius.le_lacunaryExpProduct`, `Fabius.lacunaryExpProduct_le`, for $\mathcal{E} = \text{Fabius.lacunaryExpProduct}) (level condition added after the independent check of 6 October 2026; the first wording stated the chain $2^{\binom{m}{2}}(t/2)^m \mathcal{E}(\tfrac{1}{2}) \leq \mathcal{E}(t) \leq 2^{\binom{m}{2}} t^m$ without it); the entire continuation of $D(s, a)$, $a > 0$ (`Fabius.dirichletMellinContinuation`, `Fabius.dirichletMellinContinuation_differentiable`, agreement for real $s > 1$ by `Fabius.dirichletMellinContinuation_eq`); its zeros, in the form$

`Fabius.tendsto_dirichletMellinContinuation_neg_natCast` (the evaluations `Fabius.dirichletMellinContinuation_neg_natCast` and `_zero` hold by Mathlib’s conventions $\Gamma(-r) = 0$, $0^{-1} = 0$ alone, as their docstrings warn); the dyadic equation at every complex s (`Fabius.dirichletMellinContinuation_dyadic`); the derivative bridge (`Fabius.mpLimit_eq_deriv_sub`); $\log G(\frac{1}{2}, 1) = -\frac{1}{2} \log 2$ (`Fabius.mpLimit_one_half_one`); and the evil and odious enumerators of `p1:thm:evil-odious` (`Fabius.evilEnum`, `Fabius.odiousEnum`). Not formal: the conclusion $D'(0, \frac{1}{2}) = \log 2$ of (c), which has not been stated in Lean, although its inputs are: the differentiated dyadic equation is `Fabius.thueMorseGammaLog_dyadic` at $r = 0$, $a = 1$ (where `Fabius.thueMorseGammaLog 0 a` is the derivative at $s = 0$ of the continuation and $2^0 = 1$), and the bridge and its value are `Fabius.mpLimit_eq_deriv_sub` and `Fabius.mpLimit_one_half_one` (corrected after the independent check of 6 October 2026; the first wording read “the combination in (c), which has not been written in Lean”); the identity (b); the unshifted $D(s)$, its continuation, values and derivative d ; the evaluation (5); the sector and derivative forms of flatness; and everything in Sections 3–6. This report gains no formal status from the atlas.

3 Local product models

Let $P(w) = \prod_{k \geq 1} (1 - e^{-kw})^{-1}$ be the ordinary partition product. The classical Dedekind eta transformation [12] gives the exact identity

$$P(w) = \sqrt{\frac{w}{2\pi}} \exp\left(\frac{\pi^2}{6w} - \frac{w}{24}\right) P(4\pi^2/w), \quad \Re w > 0. \quad (26)$$

Since $\Re(1/w) \geq \cos \theta / |w|$ in a fixed acute sector, the last factor is $1 + \mathcal{O}(e^{-c\theta/|w|})$ as $w \rightarrow 0$. The part indicator $(1 + \sigma\varepsilon(k))/2$ gives the exact logarithmic identity

$$\log F_\sigma(w) = \frac{1}{2} \log P(w) + \frac{\sigma}{2} H(w).$$

Combining (24) and (26), and exponentiating a flat remainder, proves

$$F_\sigma(w) = C_\sigma w^{\beta_\sigma} \exp(A/w - w/48) \{1 + \mathcal{O}_{L,\theta}(|w|^L)\} \quad (27)$$

for every fixed L . In particular $C_- = (2\pi)^{-1/4} \sqrt{Q}$ and $C_+ = (2\pi)^{-1/4} / \sqrt{Q}$, consistent with the signs in the part indicator.

For fixed $r = m + 1$, both w and rw lie in the same acute sector and tend to zero. Divide their expansions in (2); the denominator’s relative factor tends to one. This gives

$$F_{\sigma,m}(w) = r^{-\beta_\sigma} \exp(a_m/w + mw/48) \{1 + \mathcal{O}_{m,L,\theta}(|w|^L)\}. \quad (28)$$

Both d and the power of w have canceled. These local identities alone would not prove coefficient asymptotics: uniform bounds on the rest of the Cauchy contour are essential.

4 Control of every Fourier arc

We use a deliberately nonoptimal but uniform estimate. For $|z| = 1$,

$$|(1 - z)(1 - z^2)| \leq \frac{16}{3\sqrt{3}} =: B < 4. \quad (29)$$

Indeed, the left side is $8u(1 - u^2)$ for $u = |\cos(\arg(z)/2)| \in [0, 1]$; its maximum occurs at $u = 1/\sqrt{3}$. Pair adjacent factors in the length- 2^J Thue–Morse polynomial. An unpaired factor costs at most two. With

$$\alpha = \frac{\log B}{\log 4} < 1,$$

its supremum on the unit circle is $\mathcal{O}(2^{\alpha J})$. On each aligned dyadic block of length 2^J , the signs are those of the initial block multiplied by one sign; the Fourier phase adds only a unimodular factor. The binary decomposition of any prefix into such aligned blocks, with at most one of each length, gives

$$\sup_{\theta} \left| \sum_{0 \leq k < N} \varepsilon(k) e^{ik\theta} \right| = \mathcal{O}(N^\alpha). \quad (30)$$

Abel summation then implies, uniformly for real θ and $0 < t \leq 1$,

$$\sum_{k \geq 1} \varepsilon(k) e^{-kt} e^{ik\theta} = \mathcal{O}(t^{-\alpha}). \quad (31)$$

The removal of $k = 0$ costs only one. No optimal Gelfond exponent is needed.

For the allowed set write

$$U_\sigma(t, \theta) = \sum_{k \in E_\sigma} e^{-kt} (1 - \cos(k\theta)).$$

The all-integer counterpart, with $q = e^{-t}$, has the exact expression

$$U_{\text{all}}(t, \theta) = \frac{q(1+q)(1 - \cos \theta)}{(1-q)((1-q)^2 + 2q(1 - \cos \theta))}. \quad (32)$$

For $0 \leq |\theta| \leq \pi$ it increases with $|\theta|$, and at $|\theta| = t$ it is asymptotic to $1/(2t)$. By (31),

$$U_\sigma(t, \theta) = \frac{1}{2} U_{\text{all}}(t, \theta) + \mathcal{O}(t^{-\alpha}).$$

Since $\alpha < 1$, for all sufficiently small t this proves

$$U_\sigma(t, \theta) \geq c/t \quad (t \leq |\theta| \leq \pi). \quad (33)$$

This range includes every rational angle and every competing root of unity.

For $|\theta| \leq t$, restrict the sum to $1/(4t) \leq k \leq 1/(2t)$. This interval contains at least c/t complete aligned pairs $\{2j, 2j+1\}$ for small t , each containing exactly one allowed integer. For those selected terms, $e^{-kt} \geq e^{-1/2}$, $k \geq 1/(4t)$, and $|k\theta| \leq 1/2$. The elementary inequality $1 - \cos x \geq cx^2$ on this compact interval yields

$$U_\sigma(t, \theta) \geq c\theta^2/t^3 \quad (|\theta| \leq t). \quad (34)$$

The constants can be chosen positive for either sign.

For unrestricted products, the absolutely convergent logarithmic series gives

$$\log \left| \frac{F_\sigma(t - i\theta)}{F_\sigma(t)} \right| = - \sum_{k \in E_\sigma} \sum_{\ell \geq 1} \frac{e^{-\ell kt} (1 - \cos(\ell k \theta))}{\ell} \leq -U_\sigma(t, \theta). \quad (35)$$

For a capped factor, put $Z = \sum_{j=0}^m e^{-jkt}$ and $p_j = e^{-jkt}/Z$. Its normalized modulus satisfies

$$\begin{aligned} \left| \sum_{j=0}^m p_j e^{ijk\theta} \right|^2 &= 1 - 2 \sum_{j < \ell} p_j p_\ell (1 - \cos((\ell - j)k\theta)) \\ &\leq 1 - 2p_0 p_1 (1 - \cos(k\theta)). \end{aligned}$$

Here $p_0 p_1 \geq e^{-kt}/(m+1)^2$, because $Z \leq m+1$. The right side is nonnegative and $\sqrt{1-2u} \leq e^{-u}$ on $0 \leq 2u \leq 1$. Multiplication over the allowed parts proves

$$\left| \frac{F_{\sigma,m}(t-i\theta)}{F_{\sigma,m}(t)} \right| \leq \exp\left(-\frac{U_{\sigma}(t,\theta)}{(m+1)^2}\right). \quad (36)$$

In particular this argument never divides by a potentially small numerator factor near a root of unity.

Fix $0 < \delta < 1/2$ and set $h = t^{3/2-\delta} = o(t)$. Equations (33)–(36) imply that the entire region $h \leq |\theta| \leq \pi$ is suppressed by $\exp(-c_m t^{-2\delta})$ relative to the positive radial value. The saddle integral has scale that radial value times $t^{3/2}$, so the relative contribution of these arcs is at most

$$\mathcal{O}\left(t^{-3/2} e^{-c_m t^{-2\delta}}\right). \quad (37)$$

This remains smaller than every power of t . For unrestricted products the constant need not depend on m . This proves the required global arc control rather than only a local saddle approximation.

5 The completed Hankel and Bessel comparison

We now isolate a coefficient argument that applies to both products. Suppose $F(w) = \sum_{n \geq 0} f(n) e^{-nw}$ is one of the products above and, for every fixed L , has the local model

$$F(w) = C w^{\beta} e^{a/w + \lambda w} \{1 + \mathcal{O}_L(|w|^{-L})\}, \quad C > 0, \ a > 0, \ \beta, \lambda \in \mathbb{R}. \quad (38)$$

Assume the arc bound just proved. Put $N = n + \lambda$ and $t = \sqrt{a/N}$ for sufficiently large n . In the Cauchy integral under $q = e^{-w}$, the central segment is modeled by

$$\frac{C}{2\pi i} \int_{t-ih}^{t+ih} w^{\beta} e^{a/w + Nw} dw, \quad h = t^{3/2-\delta}. \quad (39)$$

For $w = t + i\theta$,

$$\Re(a/w + Nw) = \frac{2a}{t} - \frac{a\theta^2}{t(t^2 + \theta^2)}. \quad (40)$$

Since $h = o(t)$, the modulus is bounded by $e^{2a/t - c\theta^2/t^3}$ on this segment, while $|w^{\beta}| = \mathcal{O}(t^{\beta})$. Its absolute integral is therefore

$$\mathcal{O}\left(e^{2a/t} t^{\beta+3/2}\right). \quad (41)$$

Uniform flatness of the relative error in (38) gives an error

$$\mathcal{O}_L\left(e^{2a/t} t^{\beta+3/2+L}\right). \quad (42)$$

This is an absolute Gaussian majorant: no cancellation of an oscillatory integral is being used to estimate the error.

Let Γ_t be the Hankel contour coming from negative infinity just below the negative real axis to $-t$, traveling counterclockwise once around the circle $|w| = t$ from argument $-\pi$ to π , and returning above the cut to negative infinity. Expand $e^{a/w}$ along this contour, on which $|w| \geq t$. Termwise

integration is justified by uniform domination of this exponential series and by the decay of e^{Nw} at negative infinity. The reciprocal-gamma Hankel formula [13] gives

$$\frac{C}{2\pi i} \int_{\Gamma_t} w^\beta e^{a/w+Nw} dw = C \sum_{k \geq 0} \frac{a^k N^{k-\beta-1}}{k! \Gamma(k-\beta)} \quad (43)$$

$$= C \left(\frac{a}{N} \right)^{(\beta+1)/2} I_{-\beta-1}(2\sqrt{aN}) =: \mathcal{M}(n). \quad (44)$$

The second equality is the defining power series [14]. The identity is valid for every fixed real β ; zeros of $1/\Gamma(k-\beta)$ cause no problem, including at $\beta = 0$.

Here are explicit bounds completing the contour, so that (44) does not rely on a formal identification of saddle coefficients. Deform the vertical segment to the positive circular arc with the same endpoint imaginary parts, together with two horizontal connectors. Put $\rho = h/t$ and $v = \Re(w)/t$ on either connector. Then

$$\sqrt{1-\rho^2} \leq v \leq 1,$$

and the exact loss in the real exponent satisfies

$$2 - \left(\frac{v}{v^2 + \rho^2} + v \right) = \frac{(2-v)\rho^2 - v(1-v)^2}{v^2 + \rho^2} \geq \rho^2/3 \quad (45)$$

for sufficiently small ρ . Hence the connectors have exponent at most $2a/t - ch^2/t^3$. On the circle $w = te^{i\psi}$ the real exponent is exactly $2a \cos \psi/t$. Outside the central circular arc it has the same lower bound on its loss. The circle and connectors have only polynomial lengths and power factors in t , so their omitted contributions are relatively smaller than every power of t , since $h^2/t^3 = t^{-2\delta}$.

On the negative rays $w = -x \pm i0$, $x \geq t$, the absolute exponential is $e^{-a/x-Nx} \leq e^{-Nx}$. Its integral against x^β is bounded by a polynomial in t times $e^{-a/t}$, for example by scaling $x = tu$ and estimating the tail $u \geq 1$. These contributions too are smaller than every power relative to (41). All contour deformations avoid zero and the cut; the model is holomorphic in the intervening region.

Combining (37), (42), and this contour completion shows that for each fixed $L > 0$,

$$f(n) - \mathcal{M}(n) = \mathcal{O}_L(e^{2a/t} t^{\beta+3/2+L}).$$

The usual positive-argument Bessel expansion [15] gives a positive nonzero leading term of order $e^{2a/t} t^{\beta+3/2}$. Dividing by it proves the relative flat comparison

$$f(n) = \mathcal{M}(n) \{1 + \mathcal{O}_R(n^{-R})\} \quad \text{for each fixed } R > 0. \quad (46)$$

The statement concerns the coefficients of the original Euler product. No periodicity of the local Bessel model in w is assumed.

For completeness, the fixed-order Bessel expansion is

$$I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}} \left\{ \sum_{j=0}^{J-1} \frac{(-1)^j \prod_{r=1}^j (4\nu^2 - (2r-1)^2)}{j! (8z)^j} + \mathcal{O}_J(z^{-J}) \right\} \quad (z \rightarrow +\infty).$$

Its coefficients are also recorded in [16]. Substitution in (44) yields

$$f(n) = KN^{-p} e^{b\sqrt{N}} \left\{ \sum_{j=0}^{J-1} B_j(\beta, a) N^{-j/2} + \mathcal{O}_J(N^{-J/2}) \right\}, \quad (47)$$

where

$$p = \frac{\beta}{2} + \frac{3}{4}, \quad b = 2\sqrt{a}, \quad K = \frac{Ca^{\beta/2+1/4}}{2\sqrt{\pi}}. \quad (48)$$

The prefactor follows directly from $\sqrt{2\pi(2\sqrt{aN})} = 2\sqrt{\pi}(aN)^{1/4}$.

Take $(a, \beta, \lambda, C) = (A, \beta_\sigma, -1/48, C_\sigma)$ to prove Theorem 1.1. Take $(a_m, 0, m/48, r^{-\beta_\sigma})$ to prove the two coefficient statements of Theorem 1.2. Their Bessel models differ only by the factor

$$\frac{r^{-\beta_-}}{r^{-\beta_+}} = r.$$

The denominator is positive eventually by its leading equivalent; dividing the relative flat comparisons proves (14). This completes the proofs of the principal results.

6 Smooth inverses and ordinary integer thresholds

Use (a, β, λ, C) from either specialization above and define for sufficiently large real x

$$\mathcal{M}(x) = C \left(\frac{a}{x + \lambda} \right)^{(\beta+1)/2} I_{-\beta-1}(2\sqrt{a(x + \lambda)}). \quad (49)$$

This is a specified real function, not a proposed interpolation of all initial partition counts. It is eventually positive, and the Bessel derivative identity or differentiated fixed-order asymptotics give

$$\frac{\mathcal{M}'(x)}{\mathcal{M}(x)} = \frac{\sqrt{a}}{\sqrt{x + \lambda}} + \mathcal{O}(x^{-1}) > 0 \quad (50)$$

eventually. Choose a fixed tail on which positivity and strict increase hold. For large y , let $x_*(y)$ be the unique inverse of $\mathcal{M}$ on that tail. Every such choice has the same inverse once y is large enough.

With p, b, K as in (48), define

$$v_0 = -\frac{2p}{b} W_{-1} \left(-\frac{b}{2p} \left(\frac{K}{y} \right)^{1/(2p)} \right). \quad (51)$$

All our families have $p > 0$. For sufficiently large y , this is the large positive solution of

$$bv_0 - 2p \log v_0 + \log K = \log y. \quad (52)$$

The lower real Lambert branch W_{-1} , rather than W_0 , selects this solution [17]. In particular $v_0 \sim (\log y)/b$.

Theorem 6.1 (Fixed-order inverse expansions). *Let ℓ_j be the formal coefficients determined by*

$$\log \left(\sum_{j \geq 0} B_j(\beta, a) z^j \right) = \sum_{j \geq 1} \ell_j z^j. \quad (53)$$

This is a formal identity used only to finite order. Substitution of $v = \sqrt{x_(y) + \lambda} = v_0 + \sum_{j \geq 1} c_j v_0^{-j}$ in*

$$bv - 2p \log v + \log K + \sum_{j \geq 1} \ell_j v^{-j} = \log y \quad (54)$$

uniquely determines each c_j recursively. More explicitly, with a formal variable t and $\delta_{<m}(t) = \sum_{j=1}^{m-1} c_j t^j$, the recurrence is

$$c_m = -\frac{1}{b} [t^m] \left\{ -2p \log(1 + t\delta_{<m}(t)) + \sum_{j=1}^m \ell_j t^j (1 + t\delta_{<m}(t))^{-j} \right\}. \quad (55)$$

For every fixed $J \geq 1$, put $\delta_J(v_0) = \sum_{j=1}^J c_j v_0^{-j}$. Then

$$v = v_0 + \delta_J(v_0) + \mathcal{O}(v_0^{-J-1}), \quad (56)$$

$$x_*(y) = (v_0 + \delta_J(v_0))^2 - \lambda + \mathcal{O}(v_0^{-J}). \quad (57)$$

In particular, with $\ell_1 = B_1(\beta, a)$,

$$v = v_0 - \frac{\ell_1}{bv_0 - 2p} + \mathcal{O}(v_0^{-2}), \quad (58)$$

$$x_*(y) = v_0^2 - \lambda - \frac{2\ell_1}{b} + \mathcal{O}(v_0^{-1}). \quad (59)$$

Proof. The logarithm of the fixed-order Bessel expansion, evaluated at $v = \sqrt{x + \lambda}$, gives (54) to any prescribed finite order with a remainder of that order. One may initially bracket v around v_0 using the $\mathcal{O}(v_0^{-1})$ correction and the derivative $b - 2p/v$, which is bounded away from zero on the large branch; hence $v - v_0 = \mathcal{O}(v_0^{-1})$. Formal substitution is now legitimate to every fixed order. At each stage the next unknown coefficient occurs linearly with nonzero multiplier b , determining it uniquely. The residual of a fixed truncation, together with the derivative $b + \mathcal{O}(v_0^{-1})$ of the exact logarithmic Bessel model, bounds its error by the mean-value theorem. Cancellation through v_0^{-J} leaves a residual $\mathcal{O}(v_0^{-J-1})$, giving (56); squaring multiplies the error by $\mathcal{O}(v_0)$ and gives (57). The first correction solves $(b - 2p/v_0)(v - v_0) + \ell_1/v_0 = \mathcal{O}(v_0^{-2})$, proving (58); squaring gives (59). This argument never assumes convergence of the full formal series. $\square$

For the unrestricted families, equation (59) becomes

$$x_{*, -}(y) = v_0^2 + \frac{1}{48} + \frac{15}{16\pi^2} + \mathcal{O}(v_0^{-1}), \quad (60)$$

$$x_{*, +}(y) = v_0^2 + \frac{1}{48} + \frac{135}{16\pi^2} + \mathcal{O}(v_0^{-1}), \quad (61)$$

with the corresponding p and K in each v_0 . For a fixed cap,

$$x_{*, \sigma, m}(y) = v_0^2 - \frac{m}{48} + \frac{3}{16a_m} + \mathcal{O}(v_0^{-1}). \quad (62)$$

The sign affects K and hence v_0 , even though the displayed constant correction for a fixed cap is shared.

For any one of the exact counting sequences $f(n)$ define the ordinary first threshold

$$T(y) = \min\{n \geq 0 : f(n) \geq y\}. \quad (63)$$

Since $f(n) \rightarrow \infty$, this minimum exists for every real $y > 0$. For $0 < y \leq 1$, it is 0. Small values of f need not be monotone, notably for evil parts and for capped counts, so a finite computation must scan all indices in its stated range.

Proposition 6.2 (Eventual increase and rounding-aware enclosure). *Every unrestricted or fixed-cap sequence in this report is strictly increasing for sufficiently large n . If $X_J(y)$ is a finite inverse expansion with $x_*(y) - X_J(y) = \mathcal{O}((\log y)^{-q})$ for a prescribed fixed $q > 0$, there are constants $C > 0$ and y_0 such that, for $y \geq y_0$,*

$$\left\lceil X_J(y) - \frac{C}{(\log y)^q} \right\rceil \leq T(y) \leq \left\lceil X_J(y) + \frac{C}{(\log y)^q} \right\rceil. \quad (64)$$

The constant-order formulas (60)–(62) permit $q = 1$. More inverse terms permit every prescribed fixed q . Neither C nor y_0 is numerically certified here.

Proof. The relative flat comparison with $R = 1$, together with (50), implies

$$\frac{f(n+1)}{f(n)} = \frac{\mathcal{M}(n+1)}{\mathcal{M}(n)} \{1 + \mathcal{O}(n^{-1})\} = 1 + \frac{\sqrt{a}}{\sqrt{n}} + \mathcal{O}(n^{-1}) > 1$$

for large n . Equivalently, three fixed coefficient terms with relative $\mathcal{O}(n^{-3/2})$ remainder suffice; neither argument differentiates an uncontrolled discrete error.

Choose n_0 past eventual increase and the required estimates. For $y > \max_{0 \leq n < n_0} f(n)$, all initial indices are excluded, so the ordinary threshold agrees with the threshold on the monotone tail. Let $x = x_*(y)$. At integers near x , (46) gives $\log f(n) - \log \mathcal{M}(n) = \mathcal{O}_R(x^{-R})$, while (50) is bounded above and below by positive multiples of $x^{-1/2}$. Choose a sufficiently large multiple of $x^{1/2-R}$ as a positive margin. At every relevant integer below x by at least that margin, $f(n) < y$; at every integer above x by at least the margin, $f(n) \geq y$. Monotonicity extends the conclusions to all earlier and later tail indices. Thus

$$\lceil x - C_R x^{1/2-R} \rceil \leq T(y) \leq \lceil x + C_R x^{1/2-R} \rceil.$$

Since $x \asymp (\log y)^2$, take R large enough that this margin is bounded by the desired inverse-expansion uncertainty, and enlarge C to include the latter. This proves (64). $\square$

A bare ceiling of an asymptotic approximation is not justified when the approximation is close to an integer. Even an error tending to zero can change that ceiling. The exact bounded threshold routine in the companion uses integer counts and integer targets; it does not silently round a floating-point inverse. Its failure to find a threshold within the checked range means only that no index in that range attains the target.

Remark 6.3 ([write] The inverse against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. Put $v = \sqrt{x + \lambda}$, $\nu = -\beta - 1$ and $\mathcal{G}(v) = \mathcal{M}(v^2 - \lambda) = C a^{-\nu/2} v^\nu I_\nu(2\sqrt{a}v)$, so that $x_*(y) = \mathcal{G}^{-1}(y)^2 - \lambda$.

- (a) *The core is an instance of `p0:thm:lambert-core` after the change of variable $v = \sqrt{x + \lambda}$: equation (52) is that theorem's $aX + b \log X = L$ with $(a, b, X, L) := (b, -2p, v_0, \log(y/K))$. Since $p > 0$ its logarithmic coefficient is negative, and the branch rule for that sign gives the large solution, for L above the threshold L_c there, as $(-2p/b) W_{-1}(-(b/2p)e^{-L/(2p)})$, which is (51).*
- (b) *The coefficient recursion is an instance of `p0:thm:lambert-centered(2)–(4)` with $(a, b, Q) := (b, -2p, \sum_{j \geq 1} \ell_j t^j)$ over $R = \mathbb{R}$: its $\Psi(t, u) = -\frac{1}{b} [-2p \log(1 + tu) + \sum_j \ell_j t^j (1 + tu)^{-j}]$, and the coefficient of t^m in $\Psi(t, \delta_{<m}(t))$ is the right side of (55), so its c_n are the c_j of Theorem 6.1; equivalently, of `p0:thm:core-reversion` with $\Lambda = b$, $h = 0$ and $f(t, u) = -2p \log(1 + tu) + \sum_j \ell_j t^j (1 + tu)^{-j}$. The volume's `p0:eq:c-bell` gives the c_n in closed Bell-polynomial form.*

- (c) *The analytic statement (56) is an instance of `p0:thm:lambert-centered(5)`, applied to $\mathcal{G}$ on a tail $[v_1, \infty)$ with the model data $(C, a, \alpha, b, \delta, Q) := (K, b, 1, -2p, 1, \sum_j \ell_j t^j)$ of `p0:def:model`. The bound `p0:eq:model-analytic` is the logarithm of the fixed-order Bessel expansion. The derivative condition `p0:eq:model-derivative` reads $(\log \mathcal{G})'(v) = b - 2p/v + \mathcal{O}(v^{-2})$; proof: with $z = 2\sqrt{a}v$, the recurrence $I'_\nu(z) = I_{\nu-1}(z) - (\nu/z)I_\nu(z)$ (DLMF 10.29.2) gives*

$$(\log \mathcal{G})'(v) = \frac{\nu}{v} + 2\sqrt{a} \frac{I'_\nu(z)}{I_\nu(z)} = 2\sqrt{a} \frac{I_{\nu-1}(z)}{I_\nu(z)},$$

and the expansion displayed in Section 5, at the orders $\mu = \nu$ and $\nu - 1$ with two terms each, gives $I_{\nu-1}(z)/I_\nu(z) = 1 + ((4\nu^2 - 1) - (4(\nu - 1)^2 - 1))/(8z) + \mathcal{O}(z^{-2}) = 1 + (2\nu - 1)/(2z) + \mathcal{O}(z^{-2})$; hence $(\log \mathcal{G})'(v) = b + (2\nu - 1)/(2v) + \mathcal{O}(v^{-2})$, and $(2\nu - 1)/2 = -\beta - \frac{3}{2} = -2p$. $\square$ The same expansion shows that $\mathcal{G}$ is positive and strictly increasing on a tail, and it sharpens (50) to $\mathcal{M}'(x)/\mathcal{M}(x) = \sqrt{a}/\sqrt{x + \lambda} - p/(x + \lambda) + \mathcal{O}(x^{-3/2})$. Part (5) then gives $\mathcal{G}^{-1}(y) = v_0 + \sum_{n \leq J} c_n v_0^{-n} + \mathcal{O}(v_0^{-J-1})$, which is (56); (57) follows by squaring, as in the proof of Theorem 6.1. The source proves the theorem directly; its mean-value step is the argument of `p0:thm:backward-error`.

- (d) *Proposition 6.2 is not an instance as proved, only an analogue.* For y above $\max_{n < n_1} f(n)$, with n_1 past eventual increase, $T(y)$ is the integer staircase $N_*(y)$ of `p0:def:three-inverses(1)` for $A_n = f(n)$. `p0:thm:staircase(1)` writes N_* as the ceiling of the inverse of an interpolation $\mathcal{A}$ with $\mathcal{A}(n) = f(n)$; the proposition instead compares f with $\mathcal{M}$, which does not interpolate f ($\mathcal{M}(n) = f(n)$ only up to a relative $\mathcal{O}(n^{-R})$), and brackets $T(y)$ directly. Its two-ceiling form (64) plays the part of the separation condition `p0:thm:staircase(2)`: whenever the two ceilings coincide, $T(y)$ is their common value, by (64) itself.

No novelty is claimed for the inversion machinery: the source calls the Lambert- W inversion a classical tool (Section 7).

7 Attribution and the scope of the source check

The four OEIS records were independently re-fetched on 3 October 2026 from the official `oeis/oeisdata` repository and matched the frozen input records byte for byte. A067590 revision 17 and A116492 revision 18 explicitly label their leading equivalents as conjectures dated 6 July 2025. A067591 revision 6 and A116491 revision 14 contain transform descriptions but no asymptotic formula. These are the records' own revision fields, rather than a web cache's date. The report proves the stated equivalents; it does not infer historical novelty merely from their continued conjectural labels.

Allouche's seven-page primary preprint [5], published in 2021, explicitly establishes (5) and $D(0) = -1$ in the proof of Theorem 3.1, and credits Allouche–Cohen for the entire continuation. It evaluates zeta-regularized multiplicative products. Allouche's 2015 article [6], especially Sections 3.2–3.3, discusses the shifted Dirichlet continuation, Q , and the Flajolet–Martin constant. Tóth's 2022 paper [9] gives linear combinations of Thue–Morse Dirichlet series and reinforces the Allouche–Cohen attribution. None of these inspected primary texts states the restricted additive-partition coefficient equivalents proved here. The original Allouche–Cohen 1985 paper was not independently obtained; its credit is supported by the inspected primary articles, and this report reproves the needed analytic estimates rather than claiming that continuation is new.

Golafshan–Rigo’s 2026 author preprint [10] concerns multiplication of a k -ary partition series by a generalized Thue–Morse series, a digit-product k -regular coefficient sequence, and its summatory behavior. These differ from partitions whose allowed positive parts have a selected binary digit parity. Merca’s neighboring article [11] was checked only at the publisher abstract/reference level; its subscription full text was not read. It must not be included in a claim that all adjacent literature has been ruled out.

There is also genuine method and input overlap with three Formula Atlas sources in the supplied VladimirReshetnikov/ProveIt repository [18]. The bounded repository check used commit

5dacd76e280c271bb6d2d13859e167690f5c6d47.

Atlas 1, lines 855–875 and 1377–1427, gives signed ruler-weight partition/Bell identities and shifted Dirichlet continuation with real-axis flatness. Atlas 3, lines 879–919 and 1487–1533, has corresponding signed identities, continuation, nonpositive-integer zeros, and flatness. Their signed partition sums produce coefficients $+1$ or -1 , not the nonnegative counts $p_\sigma(n)$. Atlas 2, lines 832–846, 982–1045, and 2129–2213, supplies a Pascal-row statistic called a partition function, enumeration of odious and evil integers, and Laplace/shifted-Dirichlet transforms with dyadic blocks. This is additional method/input overlap, and all three are credited. Their verified Git blob IDs are recorded in the source provenance file.

Exact-ID searches returned no indexed default-branch matches for the four target sequences. Expanded searches found other odious/evil material, identified by snippets rather than exhaustively reviewed. No identical coefficient theorem was located in the primary sources and repository passages examined. This bounded negative result is not an exhaustive literature exclusion, a repository-wide originality certificate, or a historical-first claim. The eta transformation, Mellin method, Hankel–Bessel formulas, and Lambert– W inversion are classical tools used with explicit attribution.

[write] **The atlas and the OEIS records at the commit of this write, 6 October 2026.** The three atlas files cited above are archived drafts under Analysis/FabiusFunction/docs/archive/research-frontiers/; their blobs are still c927e43a, 7fee5be7 and 66c8012a, and the cited line ranges begin where the source says (“Partition and Bell-polynomial formulas” and “Shifted Dirichlet series and Mellin transforms” in Atlas 1, the Pascal-row statistic $R_u(n)$, “Evil and odious numbers” and “Laplace, Mellin, Dirichlet, and finite-product transforms” in Atlas 2, the partition formulae and “An entire shifted Dirichlet series” in Atlas 3). The atlas is now the consolidated volume named in Section 1.1; the statements overlapping this report are its p1:thm:boundary-flatness, p1:thm:Mellin-integral, p1:cor:Dirichlet-trivial-zeros, p1:thm:Dirichlet-dyadic, p1:cor:G-Dirichlet, p1:thm:Woods-Robbins and p1:thm:evil-odious, compared in Remark 2.3, which also states their formal status. The write did not map the archived partition and Pascal-row passages to the volume; as the source says, those concern signed sums or a different “partition function”, not the counts $p_\sigma(n)$. The four OEIS records were at the same revisions on 6 October 2026 (Remark 1.3). Until batch 107 placed this report, no file of the repository named the four sequences.

8 Exact companion and reproducibility

The accompanying Python code uses only the standard library for its exact checks. It fixes finite workload limits in the source rather than exposing an unbounded precision or order control. Two differently structured coefficient constructions are compared: multiplication of the part factors, and

a logarithmic-derivative Euler recurrence. To see the latter, write $P(q) = \sum_{n \geq 0} f(n)q^n$ and

$$q \frac{P'(q)}{P(q)} = \sum_{j \geq 1} b_j q^j.$$

Then $f(0) = 1$ and exact coefficient comparison gives

$$nf(n) = \sum_{j=1}^n b_j f(n-j). \quad (65)$$

For unrestricted parts,

$$b_j = \sum_{k|j, k \in E_\sigma} k.$$

For cap m and $r = m + 1$,

$$b_j = \sum_{k|j, k \in E_\sigma} k - r \mathbf{1}_{r|j} \sum_{k|(j/r), k \in E_\sigma} k. \quad (66)$$

Every division in (65) is checked for zero remainder. The coefficient algebra checks use rational arithmetic: the direct product, modified Bessel differential equation, and Gaussian saddle agree through order 6 for all three relevant values of β . The formal inverse passes both logarithmic and multiplicative composition checks through order 6. These checks are independent of floating-point approximations. Exact bounded thresholds scan the complete finite range and return a first hitting index or an explicit beyond-range result. Booleans are rejected as integer controls, and no float is silently coerced into an exact target.

The two coefficient constructions agree for both signs and caps 1, 2, 3, 5, as well as unrestricted multiplicity, through $n = 512$. The four displayed OEIS prefixes match in full, with 54, 63, 69, and 74 terms for A067590, A067591, A116492, and A116491 respectively. For example, the exact thresholds for target 100000 are 76 and 110 in the unrestricted odious and evil cases, and 170 and 186 in the distinct-part cases.

The JSON outputs separate exact statements from numerical illustrations. The exact calculations verify finite coefficient identities, selected OEIS data, and finite first thresholds. Numerical illustrations are explicitly non-certified; they neither prove an asymptotic nor supply an onset or error constant for one. In particular, none of the numerical constants displayed in this report is interval-certified. At moderate sizes, flat terms may exceed high-order algebraic truncation errors; a finite numerical fit does not determine a leading beyond-all-orders term.

The source archive contains this article, the PDF, the bounded companion and tests, deterministic reference JSON, source-provenance metadata, and reproducible local PDF/ZIP build scripts. Outputs require an existing trusted parent and an absent destination. Descriptor-pinned, no-follow, exclusive creation rejects destination or parent symlinks, existing files/directories, and parent traversal; a detected failure is not permission to overwrite. The build runs TeX without shell escape and pins its metadata. The archive uses a fixed allowlist, verified SHA-256 inputs, fixed entry order and metadata, and stored members. The release receipt records normal and optimized Python tests, two clean PDF builds, two byte-identical ZIP builds, and all-page visual review. Matching bytes are a reproducibility check, not a mathematical proof or a cross-version guarantee for arbitrary TeX installations.

[write] **The package here, and the intake reruns, 6 October 2026.** “The source archive” and “the companion” above are the delivered ones. In this report the companion program and its tests are `code/companion-companion.py` and `code/companion-test_companion.py` (delivered `companion/companion.py`,

`companion/test_companion.py`); the frozen OEIS prefixes, the reference outputs, the recorded test outputs and `verification.json` are `data/companion-*` (the reference run's SHA256SUMS is not shipped); the companion README is `companion-README.md`; the release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`; `RELEASE_RECEIPT.json` and `SOURCE_PROVENANCE.json` are in `data/`. The companion reads `oeis_prefixes.json` beside itself and the tests import it as `companion`, so the programs run only on a copy with the delivered names (the report README gives the route); the release tools expect the whole delivered package and were not run. On copies at the write (Windows, Python 3.14.4) the nine mathematical tests passed (25 s); the seven output-safety tests failed, because they need POSIX directory descriptors and ownership checks, which is also why the command line refuses to write on Windows; recomputing the two JSON payloads in process gave bytes identical to the shipped reference outputs (at placement also under -0). The release receipt's 16 of 16 passes were recorded under Python 3.12.14.

9 Conclusions and limitations

The odious unrestricted and distinct-part equivalents are proved with their exact constant expressions, and their evil counterparts follow from the same signed-product analysis. The coefficient expansions hold at every fixed algebraic order. Fixed multiplicity caps remove the signed constant and power from the local product, giving the all-order ratio $m + 1$. Uniform arc control and the completed Hankel comparison justify a relative flat error rather than merely a formal saddle series. The comparison inverse is defined as an inverse of an explicit smooth Bessel function; the ordinary integer threshold is controlled by ceilings of an enclosure.

The analysis does not provide uniform estimates for growing caps, a leading beyond-all-orders correction, interval-certified constants, effective monotonicity onsets, or numerical remainder certificates. The source search is bounded and preserves the unread-source caveats. No report was published, no OEIS entry was edited, no external party was contacted, and no repository or earlier report was modified in preparing this local deliverable.

10 Further questions and research

[write] **Added 6 October 2026, batch 107.** Following Vladimir Reshetnikov's standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The source has no questions section; its non-claims (end of Section 1.1) mark the boundaries collected below. The intake found no false claim in the source; the write found none either, and proves one negative statement (Question 1).

1. **Growing caps** (source: Theorem 1.2, "No estimate here is asserted to be uniform when m grows with n ", and the conclusions). The proof needs w and $(m+1)w$ to tend to zero together in (28), and the arc bound (36) loses the factor $(m+1)^{-2}$; all constants depend on m . [write] *The ratio law (14) is not uniform in $1 \leq m \leq n$.* Proof: no part occurs more than n times in a partition of n , so $p_{\sigma,n}(n) = p_{\sigma}(n)$, and by (9)–(10)

$$\frac{p_{-,n}(n)}{(n+1)p_{+,n}(n)} = \frac{p_{-}(n)}{(n+1)p_{+}(n)} \sim \frac{K_{-}}{K_{+}} n^{-1/2} \longrightarrow 0, \quad \frac{K_{-}}{K_{+}} = \frac{C_{-}}{C_{+}} A^{-1/2} = \frac{\sqrt{12}Q}{\pi} \approx 1.795.$$

Heuristic (not a proof): at the saddle $\Re w \asymp n^{-1/2}$, and $F_{\sigma}((m+1)w) \rightarrow 1$ when $(m+1)w \rightarrow \infty$, so the transition from the ratio $m+1$ to the unrestricted ratio $\asymp n^{1/2}$ should occur for $m \asymp \sqrt{n}$. Open: the range of $m = m(n)$ on which $p_{-,m}(n)/p_{+,m}(n) \sim m+1$, and the law in the transition

window. Missing: a local model of $F_\sigma(w)/F_\sigma((m+1)w)$ uniform in m , and an arc bound without the $(m+1)^{-2}$ loss.

2. **The residuals beyond all algebraic orders** (source: after Theorem 1.2, “We do not determine the leading residual beyond all algebraic orders”, and Section 8). Proved: the relative errors in (8), (13) and (14) are smaller than every power of n . *Observations at intake* (exact counts, mpmath models; uncertified): $p_\sigma(n)/\mathcal{M}(n) - 1 = 3.33 \cdot 10^{-8}, 8.49 \cdot 10^{-10}$ (odious) and $-4.40 \cdot 10^{-8}, -7.77 \cdot 10^{-10}$ (evil) at $n = 1000, 3000$, of opposite signs for the two signs σ ; $p_{-,m}(n)/p_{+,m}(n) - (m+1)$ at $n = 500, 1000, 1500$ is $-5.6 \cdot 10^{-5}, -1.6 \cdot 10^{-6}, -2.8 \cdot 10^{-7}$ for $m = 1$ and $+5.6 \cdot 10^{-5}, +9.8 \cdot 10^{-7}, -2.4 \cdot 10^{-6}$ for $m = 2$, so for $m = 2$ it is not of one sign in this window. These values determine no rate. Open: the leading term of each residual. A natural first step (a sketch by the write): at $q = -1$ the signed sum is again flat, since $S(w + \pi i) = (1 + e^{-w})S(2w)$ (the factor $j = 0$ of (17) becomes $1 + e^{-w}$ and the others are unchanged), so a Mellin treatment of the arc near $q = -1$, as in Section 2, looks feasible. Missing: local models at $q = -1$ and at the other roots of unity that compete, and a second saddle analysis.
3. **Effective constants and onsets** (source: Proposition 6.2, “Neither C nor y_0 is numerically certified here”, and the abstract’s “numerical onset certificate”). Exact first thresholds are known only where the companion scanned (for example $T(100000) = 76, 110, 170, 186$ for unrestricted odious, unrestricted evil, distinct odious and distinct evil parts). Missing: explicit constants in Lemma 2.1 on sectors (the formal bound $\mathcal{E}(t) \leq 2^{\binom{m}{2}} t^m$ of Remark 2.3 covers the real axis only), in the arc estimates of Section 4 and in the contour completion of Section 5; an interval value of d (from (25) or from (5)), which would also certify the digits of Kotěšovec’s decimal (Remark 1.3) (supplied from (5) by the independent check of 6 October 2026, in that remark; the error constants, onsets and threshold enclosures remain open); and from these, an explicit onset of eventual increase and certified threshold enclosures.
4. **The literature boundary** (source, Section 7). Merca’s article [11] was read at abstract level only and the Allouche–Cohen original [7] was not obtained. Open: whether either, or other work on partitions with digit-restricted parts, already contains the equivalents proved here. Missing: reading them.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 715fb82db) re-read Remarks 1.3, 2.3 and 6.3, Question 1 and the counts, with its own code. The two conjecture lines are verbatim at A067590 revision #17 and A116492 revision #18 (A067591 #6 and A116491 #14 carry no asymptotic formula), and the normalisation $N = n - \frac{1}{48}$ against Kotěšovec’s unshifted n is harmless. Counts recomputed from the definitions agree with all 54, 63, 69 and 74 listed terms and with the b-files of A067590 and A116492 ($0 \leq n \leq 10000$) and A116491 ($0 \leq n \leq 1000$); the relative Bessel residuals quoted in Question 2 were reproduced. The bridge (b) of Remark 2.3 was re-proved and checked numerically (mpmath, 40 digits): $\partial_s D(s, \frac{1}{2})|_{s=0} = \log 2$ to 30 digits, the second derivative $1.156203063244203004818459767$ to 40 digits, $D(2, \frac{1}{2}) = -3D(2) = 3.4657672610904246175$, and $|D(2\pi i / \log 2, \frac{1}{2})| = 4.5 \times 10^{-37}$ (control at $1 + 2\pi i / \log 2$: 2.34). Each Lean declaration named in (d) was re-read and states what is claimed. Remark 6.3 was confirmed item by item (the W_{-1} residuals are below 10^{-29}). For Question 1, $K_-/K_+ = \sqrt{12} Q/\pi = 1.7953034517754646109$, and $\sqrt{n} p_-(n)/((n+1)p_+(n)) = 1.9497, 1.8334, 1.8175$ at $n = 100, 1000, 3000$. No mathematical error was found. Three precision fixes to Remark 2.3 are made, each with a dated note keeping the first wording: the dyadic equation of the atlas is stated for $\Re s > 0$ only, so differentiating it at $s = 0$ cites its entire form `p1:eq:Dirichlet-dyadic-entire` (Fabi.us.dirichletMellinContinuation_dyadic); the lower bound `Fabi.us.le_lacunaryExpProduct` needs the level condition $\frac{1}{2} \leq 2^m t \leq 2$; and the differentiation step of (c) is already formal, as `Fabi.us.thueMorseGammaLog_dyadic` at $r = 0, a = 1$, so only the conclusion

$D'(0, \frac{1}{2}) = \log 2$ is not stated in Lean. Added: the interval enclosure of $\log Q$ in Remark 1.3, which certifies the nine printed digits of c given (5).

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